

Ethernet Standard and Specification

Question 1: Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Answer:

An RJ-45 connector has 8 pins numbered 1 to 8 from left to right when viewed from the contact side with the clip facing away. The diagram below shows the pin layout with the wire colour of each pin for both wiring standards. The two standards differ only in the positions of the green and orange pairs (pins 1, 2, 3 and 6).



W/ = White with stripe of that colour (striped wire). Pin 1 is on the left.

Pins 1-2 = TX pair, pins 3-6 = RX pair (used by 10BASE-T and 100BASE-TX).

Figure 1: RJ-45 pin layout with T568A and T568B colour codes

Question 2: Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Answer:

Specification	10BASE-T	100BASE-TX	1000BASE-T
IEEE standard	802.3i	802.3u	802.3ab
Common name	Ethernet	Fast Ethernet	Gigabit Ethernet
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	UTP Category 3 or higher	UTP Category 5 or higher	UTP Category 5e or higher (Cat 6 preferred)
Pairs used	2 pairs (1-2 and 3-6)	2 pairs (1-2 and 3-6)	All 4 pairs (bidirectional)
Maximum cable length	100 metres	100 metres	100 metres

Specification	10BASE-T	100BASE-TX	1000BASE-T
Connector type	RJ-45	RJ-45	RJ-45
Signal encoding	Manchester	4B/5B with MLT-3	PAM-5

Question 3: Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Answer:

I would use a **crossover cable**.

A laptop network card (NIC) is a DTE-type device: it transmits on pins 1 and 2 and receives on pins 3 and 6. In a straight-through cable every pin connects to the same pin at the other end (1 to 1, 2 to 2, 3 to 3, 6 to 6). If two laptops are joined with a straight-through cable, the transmit pins of laptop A reach the transmit pins of laptop B, and the receive pins of both laptops are connected to each other. Both laptops would be "speaking" on the same pair and nobody would be "listening", so no communication is possible.

A crossover cable is wired with T568A at one end and T568B at the other, which swaps the transmit and receive pairs. The transmit pair of one laptop is connected to the receive pair of the other:

Laptop A pin	Function	Connects to Laptop B pin	Function
1	TX+	3	RX+
2	TX-	6	RX-
3	RX+	1	TX+
6	RX-	2	TX-

For a Gigabit link (1000BASE-T), a full crossover cable also swaps pins 4-5 with pins 7-8. Note that most modern laptops support Auto MDI/MDI-X, which corrects the pairs automatically, so a straight-through cable would also work with them. However, according to the wiring standard, a crossover cable is the correct choice for a direct laptop-to-laptop (same device type) connection.

Question 4: List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Answer:

S. No.	Device	Where used	Likely Ethernet standard
1	Desktop computer / laptop (network card)	College computer lab and home	1000BASE-T (Gigabit Ethernet)
2	Wi-Fi router / network switch (LAN ports)	Home and college network rooms	1000BASE-T (older models: 100BASE-TX)
3	Network printer / IP CCTV camera	College office and campus security	100BASE-TX (Fast Ethernet)